

local-1777884295150

Comprehensive Hardware Design Analysis

Revision 1 | Generated 04/05/2026

Executive Summary

Generated

Project ID	local-1777884295150
Industry Domain	commercial_hvac
Engine Version	PDF Engine v2.1 (Full Detail)
Target Quantity	500 units annually

ARCHITECTURE AT A GLANCE

MODULES	BOM ROWS	IDENTIFIED RISKS
8	53	24

ECONOMICS

EST. UNIT COST	TARGET CEILING	HEADROOM
£5,926.50	None	N/A

TOTAL NRE

£4,800.00

UNIT + NRE AMORTISED

£5,936.10

Contents

- 1. Brief & Requirements**
- 2. Regulatory & Compliance Posture**
- 3. Sizing & Spatial Optimisation**
- 4. System Modules & Architecture**
- 5. Bill of Materials (BOM)**
- 6. Cost Waterfall & Economics**
- 7. Risks Register (FMEA)**
- 8. Supplier Shortlist & Sourcing**
- 9. Audit Log**
- 10. Source Attribution**
- 11. Quality Scores**

1. Brief & Requirements

1.1 Overview & Context

Subject / Use Case

Primary heating and potential hot water generation for small commercial buildings (offices, retail, schools) in the UK operating in standard winter climates.

Full Synthesis Report

1. Executive Summary

The founder's brief outlines the development of a compact commercial air-source heat pump (ASHP) designed for small commercial buildings. The core specifications include a 30kW thermal output capacity, a Coefficient of Performance (COP) of e 3.5 at standard A2/W35 test conditions, and the use of R290 (propane) as the primary refrigerant. Mechanical constraints mandate a stainless steel heat exchanger, a scroll compressor, and an axial fan within an outdoor enclosure no larger than 1200mm W x 800mm D x 1400mm H. The system targets a rigorous acoustic threshold of < 65dB(A) at 1 metre, and an aggressive unit manufacturing cost of £4,500 at an annual production volume of 500 units.

A fundamental pivot is required regarding the "indoor unit + outdoor unit" architecture. Standard direct expansion (DX) split systems utilizing R290 are heavily constrained by indoor lower flammability limits (LFL), dictated by BS EN 378 and IEC 60335-2-40. For a 30kW unit, the R290 charge will likely exceed 2.0kg. Routing highly flammable A3 refrigerant piping into an indoor space requires exhaustive safety mechanisms, ATEX-rated indoor ventilation, and complex room-size-dependent calculations that would preclude standardized small-commercial deployment. Therefore, this dossier re-engineers the "split system" into a hermetically sealed R290 outdoor monobloc, paired with an indoor "Hydronic Box" (comprising circulating pumps, expansion vessel, backup immersion heaters, and smart controls). This guarantees absolute compliance with UK safety codes, secures Microgeneration Certification Scheme (MCS) eligibility, and standardizes installation.

2. Thermodynamic & Refrigerant Analysis: R290

R290 (Propane, C₃H₈) is deployed due to its ultra-low Global Warming Potential (GWP = 3), zero Ozone Depletion Potential (ODP), and superior thermodynamic efficiency. It is classified as an A3 (highly flammable, non-toxic) refrigerant.

1. Physical Properties:

- Boiling Point: -42.1°C at atmospheric pressure.
- Critical Temperature: 96.7°C, allowing excellent performance at high condensing temperatures without operating close to the transcritical boundary.
- Latent Heat of Vaporization: ~425 kJ/kg, significantly higher than R410A (~210 kJ/kg) or R32 (~270 kJ/kg). This high latent heat means R290 requires a substantially lower mass flow rate to achieve the same thermal duty, reducing compressor pumping work and pressure drops inside the heat exchangers.

2. Cycle Analysis at A2/W35:

To achieve a 30kW thermal output and a COP e 3.5 at A2/W35 (Ambient Air 2°C / Leaving Water 35°C), the internal pressures and temperatures are modeled as follows:

- Evaporating Temperature: ~ -4°C to -6°C (Delta T of 6K-8K across the evaporator coil to maintain compact dimensions).
- Condensing Temperature: ~ 38°C (Delta T of 3K across the brazed plate heat exchanger).
- Superheat target: 5K, Subcooling target: 5K.
- The pressure ratio (P_c/P_e) at these conditions for R290 is exceptionally low (approx. 3.0), enabling an isentropic compressor efficiency of 70-74%. The theoretical Carnot COP at these temperatures is roughly 7.0. Achieving a baseline system COP of 3.5 gives a Second Law efficiency of 50%, which is well within the capabilities of a modern variable-speed BLDC scroll compressor and Electronic Expansion Valve (EEV) architecture.

3. Component Engineering & Sizing

3.1. Compressor

The requirement specifies a scroll compressor. An inverter-driven (DC brushless) scroll compressor optimized for R290 must be selected. Emerson Copeland (e.g., ZHV series) or Bitzer offer propane-specific scrolls equipped with robust Polyolester (POE) or Polyalkylene Glycol (PAG) oil management systems. At 30kW nominal capacity (A2/W35), factoring an enthalpy rise through the condenser, the mass flow rate is roughly 0.08 kg/s (288 kg/h). The required swept volume at max RPM (usually 90-120Hz) will be roughly 18 - 22 m³/h. An acoustic jacket, fabricated from high-density mass-loaded vinyl and mineral wool, is mandatory to suppress the 500-2000Hz frequency bands radiating from the scroll shell.

3.2. Air-side Heat Exchanger (Evaporator)

The outdoor envelope constraint is strict: 1200mm (W) x 800mm (D) x 1400mm (H). The maximum theoretical face area for an evaporator coil placed on multiple sides (e.g., U-shape or L-shape configuration) must be maximized. Given an internal frame layout, an L-shaped coil measuring approx. 1100mm (H) x 1800mm (L) yields ~1.98 m² of face area. To extract ~24kW of heat from 2°C air (the remainder of the 30kW is compressor input heat), a massive volume of air is required. Airflow must be maintained near 1.5 - 2.0 m/s face velocity to prevent excessive pressure drops and fan noise. This dictates an airflow of ~3.5 m³/s (12,600 m³/h). The coil should utilize 7mm copper tubes with louvered hydrophilic aluminium fins to accelerate condensate drainage and delay frost formation (fin pitch 2.1mm).

3.3. Water-side Heat Exchanger (Condenser)

The brief mandates a stainless steel heat exchanger. An asymmetric Brazed Plate Heat Exchanger (BPHE), constructed from 316L stainless steel brazed with 99% pure copper (or AlfaNova 100% stainless fusion brazing for high purity/durability) is the optimal solution. To transfer 30kW at W35 with a Delta T of 5K on the water side, the water mass flow rate must be:

$$Q = m \cdot C_p \cdot \Delta T \rightarrow 30\text{kW} = m \cdot 4.18 \text{ kJ/kg.K} \cdot 5\text{K} \rightarrow m = 1.43 \text{ kg/s (approx. 5.1 m}^3\text{/h)}.$$

A unit comparable to the Alfa Laval CB76 or CB112 ranges with 60 to 80 plates will suffice to keep water-side pressure drop below 25 kPa, permitting the use of standard high-efficiency EC circulation pumps (e.g., Wilo Stratos MAXO).

3.4. Axial Fan Strategy

Meeting < 65dB(A) at 1 metre for a 30kW unit moving 12,600 m³/h is the paramount engineering challenge of this brief. Conventional AC fans will easily exceed 72dB at full load. The solution lies in large-diameter, low-RPM EC (Electronically Commutated) axial fans with biometric/swept blade designs (e.g., Ziehl-Abegg OWlet or ebm-papst AxiBlade). Given the 1200mm width constraint, a dual-fan vertical stack (2 x 500mm fans) or a single massive 710mm fan can be employed. A single 710mm EC fan operating at roughly 750 RPM can move the necessary air while remaining below 62dB(A) Sound Power Level (SWL), yielding a Sound Pressure Level (SPL) well below 65dB(A) at 1m in free-field conditions. Passive acoustic damping (e.g., foam baffles on the compressor compartment, rubber-in-shear vibration isolators on the fan chassis) is critical.

4. Operating Parameters & Control Strategy

Advanced logic must control the Electronic Expansion Valve (EEV) and inverter compressor simultaneously to maintain a target evaporator superheat of 4-6K while modulating compressor speed to hit the exact thermal load.

Frost management is a core efficiency metric. Standard ASHPs utilize basic time-and-temperature algorithms which often

trigger unnecessary reverse-cycle defrosts. For UK commercial application (where A2/W35 operations occur frequently in high-humidity winter mornings), intelligent defrosting via evaluating the divergence between ambient air temperature, coil temperature, and fan current (as frost builds, air resistance increases, lowering EC fan motor current) ensures defrosting only when strictly necessary, safeguarding the Seasonal COP (SCOP) essential for MCS certification.

5. Safety Architecture for R290 (ATEX and DSEAR)

Because R290 is flammable, the electrical compartment within the outdoor unit must be isolated from the refrigeration circuit. Components that could spark (contactors, standard relays) must be housed in a sealed IP65-rated control box, ideally maintained under slight positive pressure or situated entirely upstream of any potential refrigerant airflow.

By ensuring the "Indoor Unit" is exclusively a hydronic module, zero A3 refrigerant enters the building footprint. All compliance with BS EN 378 dictates focus on the outdoor installation site: ensuring the outdoor unit is situated away from basement lightwells, drains, or ventilation intakes, as R290 is heavier than air (density at standard state is 1.88 kg/m³, roughly 1.5 times that of air).

6. Manufacturing Economics and BOM Analysis

The target unit cost is £4,500 based on an annual run of 500 units. A 500/year volume is relatively low for bespoke tooling; therefore, Commercial-Off-The-Shelf (COTS) components and standard sheet-metal-brake fabrication (avoiding deep-drawn stamping dies) must be leveraged vigorously.

Estimated Bill of Materials (BOM) breakdown at 500 units/year:

- R290 Inverter Scroll Compressor: £750
- Inverter Drive / BLDC Controller: £450
- Outdoor Air Evaporator Coil (Custom dimensioned): £550
- EC Axial Fan Assembly (e.g., 710mm Ziehl-Abegg): £400
- SS316 Braze Plate Heat Exchanger: £300
- Primary EEV, 4-Way Valve, Pressure Transducers: £250
- Structural Enclosure (Powder-coated Galvanized Steel): £500
- Hydronic Assembly Components (Pump, flow sensors, expansion): £350
- Main PCB, Sensors, HMI panel: £250
- Acoustic dampening materials: £100
- Refrigerant, piping, valving, brazing material: £150
- Assembly Labor, Helium Leak & EOL Testing: £300
- Freight, Packaging, Contingency: £150

Total Estimated Direct Unit Cost: £4,500.

Achieving this demands tight supply chain consolidation, likely procuring the customized evaporator coil, enclosure chassis, and PCBs from Tier 1 contract manufacturers in Southeast Asia, while sourcing highly sensitive mechanics (compressor, BPHE, flow valves) from established European channels to guarantee MCS reliability.

7. Regulatory Certification (MCS and UK Market)

To access UK commercial ESG funding and government boiler upgrade incentives, the product demands MCS certification (Microgeneration Certification Scheme) under standard MIS 3005-I. Achieving this status requires rigorous independent laboratory testing:

- EN 14511 Testing: Validating the 30kW output and COP e 3.5 at various partial loads.
 - EN 14825 Testing: Calculating the SCOP (Seasonal Coefficient of Performance) for the ErP (Energy-related Products) label.
- A 30kW commercial R290 unit should aim for an ErP rating of A++ or A+++ (SCOP > 4.2 at low temperature W35 average

climate).

- EN 12102 Acoustic Testing: Verifying the < 65dB Sound Power/Pressure claims in an anechoic chamber.
- UK DSEAR: Dangerous Substances and Explosive Atmospheres Regulations compliance file for handling R290 both at the factory assembly line and field servicing.

8. Conclusion and Engineering Roadmap

The founder's brief is fundamentally viable but resides on the bleeding edge of current volumetric power density and acoustic limitations for A3 refrigerant ASHPs. The critical pathway to success relies on pivoting the split-system definition to a monobloc-hydrobox paradigm to circumvent indoor flammability limits. The mechanical engineering focus must rapidly shift into advanced Computational Fluid Dynamics (CFD) for the evaporator coil optimization and acoustic barrier specification, as hitting 65dB at 1m with a 30kW thermal capacity is severe. Procuring prototype hardware from Bitzer/Copeland and ebm-papst will dictate the timeline. The £4,500 BOM target is achievable but restricted, leaving little margin for bespoke electronic drive R&D, mandating the use of integrated third-party heat pump drive platforms (such as Carel or Eliwell) embedded into the proprietary sheet-metal casing.

1.2 Market & Competitors

Competitor Landscape

Mitsubishi Electric

Product	Ecodan CAHV Air Source Heat Pump
Pricing Strategy	Est. £7,500 - £9,000 unit cost
Technical Specs	Cascadable commercial system, primarily R410A / R454B transitional range, output up to 43kW, hermetic scroll.

STRENGTHS

Phenomenal brand trust, highly integrated cascade controls, extremely robust in UK weather environments.

WEAKNESSES

Higher unit cost, legacy refrigerant phase-down vulnerabilities, larger footprint.

DIFFERENTIATION ANGLE

Significantly lower capital cost (£4,500 BOM constraint implies ultra-competitive wholesale), much lower GWP (3 vs >2000), smaller single-chassis footprint.

Vaillant

Product	aroTHERM plus 15kW (Cascaded x 2)
Pricing Strategy	Est. £4,000 per 15kW unit (Total ~£8,000 for 30kW capability)
Technical Specs	R290 refrigerant monobloc, ~7kW to 15kW units cascaded to 30kW, COP 3.6 to 4.8 depending on condition, MCS certified.

STRENGTHS

Market leader in UK R290 adoption, exceptionally quiet (approx 53dB(A) Sound Power per unit), beautifully refined sheet metal.

WEAKNESSES

Achieving 30kW requires purchasing and installing two distinct physical outdoor units and a complex cascaded controller.

DIFFERENTIATION ANGLE

Consolidating 30kW into a single outdoor frame (1200x800) halves installation footprint and significantly cuts capital costs compared to twin 15kW modules.

Clivet

Product	Thunder R290 30.2 kW
Pricing Strategy	Est. £5,500 - £6,500
Technical Specs	R290 Full-DC inverter heat pump, 30.2kW nominal heating, dual EC axial fans, integrated hydronic pump.

STRENGTHS

Dedicated purpose-built commercial R290 unit, high flow temperatures (up to 75C), sophisticated Italian design.

WEAKNESSES

Heavy (over 250kg), high structural cost burden, post-Brexit import complexities for spare parts.

DIFFERENTIATION ANGLE

Aggressive localized UK assembly and specific acoustic tuning. Leveraging single massive 710mm fan to out-perform dual fan acoustics on equivalent capacity.

1.3 Engineering Constraints

Core Targets

Unit Cost Ceiling	£0.00
Max Mass	-
Target Process	Air-to-water thermodynamic vapor compression cycle heavily optimized for variable load tracking and zero-flammability risk indoors.
Target Material	R290 Refrigerant, 316L Stainless Steel Brazed Plate Heat Exchanger, Powder-coated Z275 Galvanized Steel Enclosure, Copper/Hydrophilic Aluminum Fin-and-Tube Evaporator.
Tolerance Target	Strict acoustic compliance < 65dB at 1m. Heat duty +/- 2% of 30kW at A2/W35. Enclosure precisely under 1200x800x1400mm.
Production Quantity	500 units annually

Compliance & Standards Notes

MCS certification is mandatory. Must mitigate highly flammable A3 refrigerant characteristics strictly containing all re- frigerant outdoors in a sealed monobloc architecture interfacing with an indoor hydronic-only unit.

2.1 BS EN 14511

Air conditioners, liquid chilling packages and heat pumps for space heating and cooling

Status: not-started

Owner: Thermodynamics Engineer

Standard Summary

Standard for testing and rating of heat pumps defining A2/W35 test conditions.

Applicability Assessment

Mandatory for verifying 30kW output and COP 3.5.

Engineering Design Impact

Requires rigorous sensor placement and heat exchanger sizing to meet published claims.

EVIDENCE REQUIRED

Independent lab test report.

GAP ACTION

Book climatic chamber for alpha prototype testing.

2.2 BS EN 14825

Testing and rating at part load conditions and calculation of seasonal performance

Status: not-started

Owner: Compliance Engineer

Standard Summary

Standard procedure to calculate the Seasonal Coefficient of Performance (SCOP).

Applicability Assessment

Mandatory for UK ErP labeling (energy-related products).

Engineering Design Impact

Demands high-efficiency low-RPM capability from the inverter scroll and EC fans.

EVIDENCE REQUIRED

Multi-point testing log and SCOP calculation sheet.

GAP ACTION

Write part-load PID control algorithm.

2.3 BS EN 378

Refrigerating systems and heat pumps. Safety and environmental requirements

Status: not-started

Owner: Chief Mechanical Engineer

Standard Summary

Governs the maximum allowable charge limits and mitigation strategies for highly flammable (A3) refrigerants.

Applicability Assessment

Strictly governs the physical routing and architecture of the R290 circuit.

Engineering Design Impact

Forces abandonment of indoor DX R290 unit in favor of an outdoor sealed monobloc with indoor hydronic box.

EVIDENCE REQUIRED

LFL calculation assessment and leak simulation report.

GAP ACTION

Audit enclosure design for vapor extraction.

2.4 IEC 60335-2-40

Particular requirements for electrical heat pumps, air-conditioners and dehumidifiers

Status: not-started

Owner: Electrical Engineer

Standard Summary

Electrical safety code handling the intersection of high voltage electronics and flammable gasses.

Applicability Assessment

Mandatory for CE/UKCA marking of the product.

Engineering Design Impact

Requires sealed IP65 flame-proof enclosures for all contactors internally switching near R290.

EVIDENCE REQUIRED

Certification body safety mark (e.g. TUV, BSI).

GAP ACTION

Source ATEX-rated relays for control board.

2.5 MIS 3005-I

Microgeneration Installation Standard for Heat Pumps

Status: not-started

Owner: Product Manager

Standard Summary

UK specific MCS standard designating eligibility requirements for heat pump installations.

Applicability Assessment

Crucial requirement for the commercial success of the product in the UK.

Engineering Design Impact

Product data must seamlessly integrate into MCS heat loss and sound calculators.

EVIDENCE REQUIRED

MCS listing on the official product database.

GAP ACTION

Submit technical dossier to an MCS certification body.

2.6 ISO 3744

Acoustics - Determination of sound power levels and sound energy levels of noise sources using sound pressure

Status: not-started

Owner: Acoustics/NVH Engineer

Standard Summary

Methodology to measure sound output across an enveloping hemisphere.

Applicability Assessment

Required to legally print '< 65dB(A)' on the rating plate and marketing materials.

Engineering Design Impact

Dictates chassis rigidity, acoustic foam thickness, and fan blade profile design.

EVIDENCE REQUIRED

Acoustic laboratory certification report.

GAP ACTION

Conduct CFD and acoustic isolation material prototyping.

2.7 PED 2014/68/EU

Pressure Equipment Directive

Status: not-started

Owner: Manufacturing Engineer

Standard Summary

EU and UK regulatory code for pressurized systems (retained in UK law).

Applicability Assessment

A3 refrigerants at high pressures classify piping and vessels under high hazard categories.

Engineering Design Impact

Requires strict control over copper tube wall thickness and certified brazing protocols.

EVIDENCE REQUIRED

Notified body inspection material certificates (Type Examination).

GAP ACTION

Establish EN 13134 brazing procedure qualifications.

2.8 DSEAR 2002

Dangerous Substances and Explosive Atmospheres Regulations

Status: not-started

Owner: Health & Safety Officer

Standard Summary

UK framework protecting factory workers and installers from explosion risks associated with R290.

Applicability Assessment

Governs both the assembly line at the 500 units/year factory and field maintenance manuals.

Engineering Design Impact

Determines warning label placements, factory ventilation requirements, and maintenance access port designs.

EVIDENCE REQUIRED

DSEAR risk assessment file.

GAP ACTION

Design custom leak-proof R290 charging rigs for assembly line.

3. Sizing & Spatial Optimisation

INFEASIBLE DESIGN ALLOCATION

The current sizing constraints are not satisfied. Module requirements exceed the available envelope.

System Envelope

KIND	RULES DOMAIN	FEASIBLE
generic_room	commercial_hvac	No

Internal Dimensions (W×D×H)	5000 × 5000 × 3000 mm
Total Internal Floor Area	25 m²
Total Internal Volume	75 m³
Available Floor Budget	21.25 m²

Module Dimension Allocations

MODULE	W × D × H (MM)	AREA M²	MOUNT	SCALE
structural_enclosure	2236 × 2236 × 2400	5	floor	mass_proportional
compressor_assembly	2236 × 2236 × 2400	5	floor	mass_proportional
evaporator_coil	2236 × 2236 × 2400	5	floor	mass_proportional
axial_fan_assembly	2236 × 2236 × 2400	5	floor	mass_proportional
condenser_bphe	2236 × 2236 × 2400	5	floor	mass_proportional
refrigerant_valving_circuit	2236 × 2236 × 2400	5	floor	mass_proportional
electrical_control_box	2236 × 2236 × 2400	5	floor	mass_proportional

indoor_hydronic_box	2236 × 2236 × 2400	5	floor	mass_proportional
---------------------	--------------------	---	-------	-------------------

Identified Conflicts

- Total module footprint (40.0m2) exceeds envelope budget (21.3m2)

Optimisation Recommendations

- Consider a larger envelope
- Optimize module footprint

4.1 Structural & Acoustic Enclosure

Lead Time: 6 weeks

Status: draft

Module Purpose

Provides the physical framework, weatherproofing, and primary acoustic barrier for the outdoor monobloc.

Why it matters to the system

Without superior acoustic compartmentalization and rigid structural resonance suppression, achieving the severe < 65dB(A) at 1m threshold is physically impossible.

Detailed Technical Description

The structural enclosure forms the highly rigid 1200mm (W) x 800mm (D) x 1400mm (H) framework necessary to house the heat pump's dynamic components. Fabricated primarily from powder-coated Z275 galvanized steel via standard sheet-metal-brake bending, it minimizes deep-drawn stamping to accommodate the stringent £4,500 BOM constraint for a low 500-unit annual volume.

Critically, this module is not just a skin but a primary acoustic and safety barrier. It incorporates dedicated mass-loaded foam baffles to attenuate compressor drone and features a compartmentalized layout that strictly segregates the electrical housing from the refrigeration circuit, a primary requirement for safe R290 (propane) deployment. The design structurally suppresses harmonic resonance from the massive 710mm fan and compressor via rubber-in-shear mounts, enabling compliance with strict ISO 3744 acoustic parameters.

SYSTEM INPUTS	SYSTEM OUTPUTS
- Ambient air (intake) - Physical foundation support	- Exhaust air - Vibration-damped structural load

4.1.2 Specifications & Risk Profile

Expected Key Parts

- Powder-coated Z275 galvanized steel panels
- Welded skeletal chassis
- Mass-loaded acoustic foam baffles
- Rubber-in-shear chassis mounts
- ATEX ventilation grilles

Identified Failure Modes

- Galvanic corrosion at fastener points leading to structural fatigue
- Acoustic panel delamination causing blockage of airflow routing
- Chassis resonance matching the 750 RPM frequency of the axial fan, amplifying structural noise

Open Questions

- Precise stiffness required to decouple multi-axis compressor vibration
- Long-term UV resistance of the powder-coated finish in coastal UK climates

4.1.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
structural_enclosure-001	Powder-coated Z275 galvanized steel panels	COTS	2kg	£40.00
structural_enclosure-002	Welded skeletal chassis	COTS	2kg	£120.00
structural_enclosure-003	Mass-loaded acoustic foam baffles	COTS	2kg	£40.00
structural_enclosure-004	Rubber-in-shear chassis mounts	COTS	2kg	£120.00
structural_enclosure-005	ATEX ventilation grilles	COTS	2kg	£40.00
structural_enclosure-fasteners	M8 Stainless Steel Assembly Fasteners Kit	COTS	1.2kg	£15.00
structural_enclosure-gaskets	EPDM Weatherproof Panel Gasketing	COTS	0.8kg	£12.50
structural_enclosure-labels	ATEX / Flammable Gas Warning Labels	COTS	0.05kg	£3.00
Module Total Cost:				£390.50

4.1.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The sheet metal brake-bending approach aligns perfectly with our low-volume constraints, but welding Z275 galvanized steel poses significant corrosion and manufacturing health risks. Additionally, mechanical retention mechanisms must be mandated for the acoustic foam to prevent delamination in fluctuating outdoor climates.

Identified Issues

MANUFACTURABILITY

Welding Z275 galvanized steel vaporizes the zinc coating at the weld seams, creating toxic fumes for operators and leaving the joint highly susceptible to corrosion if not perfectly pre-treated before powder coating.

Suggestion: Transition to a fully bolted, blind-riveted, or Tog-L-Loc (clinched) chassis assembly to eliminate welding. If welding is mandatory, enforce strict SOPs for local exhaust ventilation and post-weld zinc-rich priming.

MATERIALS

Mass-loaded acoustic foam is prone to delamination when relying solely on adhesives in outdoor thermal and humidity cycles. Fallen foam can block airflow, causing catastrophic compressor failure.

Suggestion: Implement mechanical fasteners, such as capacitor discharge (CD) weld pins with domed retaining washers, to physically secure the acoustic baffles.

RISK

Galvanic corrosion at fastener points is a noted failure mode. Using stainless steel fasteners directly against Z275 or powder-coated steel will create a galvanic cell when moisture is present.

Suggestion: Specify zinc-flake coated fasteners (e.g., Geomet/Dacromet) or use nylon isolation washers to prevent dissimilar metal contact.

ENGINEERING

Chassis resonance at 750 RPM (12.5 Hz) from the fan is difficult to damp. Standard rubber mounts may harden in sub-zero outdoor temperatures, shifting their natural frequency and failing to isolate the 12.5 Hz vibration.

Suggestion: Ensure the rubber-in-shear mounts are molded from a weather- and temperature-stable elastomer like EPDM or silicone, and dynamically test them at -20°C.

RISK

Strict segregation of electrical housings from the R290 refrigeration circuit requires perfectly sealed penetrations to maintain ATEX compliance.

Suggestion: Specify ATEX-certified, gas-tight cable glands for any wire routing between the electrical and compressor compartments.

Strategic Recommendations

- Redesign the skeletal chassis to utilise rivet nuts or structural blind rivets instead of welding to preserve the Z275 zinc coating.
- Draft a BOM update to include mechanical retention pins for all acoustic foam panels.
- Standardise fastener coatings to zinc-flake (e.g., Geomet 500) across the entire enclosure to eliminate galvanic corrosion risks.
- Validate the rubber-in-shear chassis mounts in an environmental chamber at -20°C to ensure 12.5 Hz transmissibility remains within spec.
- Verify that the physical barrier between the R290 zone and electrical zone passes a positive pressure leak test in the assembly line.

4.2 Inverter Scroll Compressor Assembly

Lead Time: 14 weeks

Status: draft

Module Purpose

Drives the thermodynamic cycle by compressing low-pressure R290 vapor into high-pressure, high-temperature gas.

Why it matters to the system

The efficiency of this module dictates the system's ability to achieve a COP e 3.5 at A2/W35, necessary for MCS certification and ErP A+++ rating.

Detailed Technical Description

The compressor assembly represents the thermodynamic heart of the heat pump. Utilizing a high-efficiency Brushless DC (BLDC) scroll compressor specialized for explosive A3 refrigerants (such as the Emerson Copeland ZHV or Bitzer equivalent), it takes advantage of R290's high latent heat of vaporization (425 kJ/kg). By operating at an exceptionally low pressure ratio (P_c/P_e of approx 3.0), the compressor attains an isentropic efficiency of over 70%, driving a mass flow rate of 0.08 kg/s to meet the 30kW target.

To manage the inherent Noise, Vibration, and Harshness (NVH) created by a scroll set pumping 18-22 m³/h of swept volume at 120Hz, an intensive dual-stage dampening approach is employed. The compressor is swaddled in a custom mass-loaded vinyl and mineral wool acoustic jacket to suppress 500-2000Hz frequency bands, while specialized rubber mounts isolate low-frequency cyclic thumping from reaching the primary structural enclosure.

SYSTEM INPUTS	SYSTEM OUTPUTS
- Low-pressure R290 vapor (-4°C) - High-voltage 3-phase DC (from inverter)	- High-pressure R290 vapor (approx 38°C condensing equivalent) - Thermal radiation - Mechanically transmitted vibration

4.2.2 Specifications & Risk Profile

Expected Key Parts

- BLDC R290 Scroll Compressor
- Polyolester (POE) lubricating oil
- Mass-loaded vinyl acoustic jacket
- Vibration-isolating rubber feet

Identified Failure Modes

- Liquid slugging destroying the scroll involutes during cold starts
- POE oil breakdown due to excessive discharge temperatures
- Loss of variable frequency drive phase leading to motor stall/burnout

Open Questions

- Optimal acoustic jacket thickness to balance noise reduction with motor thermal shedding
- Exact volumetric efficiency curve on partial load tracking below 30% speed

4.2.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
compressor_assembly-001	BLDC R290 Scroll Compressor	COTS	2.5kg	£800.00
compressor_assembly-002	Polyolester (POE) lubricating oil	COTS	2.5kg	£40.00
compressor_assembly-003	Mass-loaded vinyl acoustic jacket	COTS	2.5kg	£40.00
compressor_assembly-004	Vibration-isolating rubber feet	COTS	2.5kg	£40.00
compressor_assembly-hardware	M10 Compressor Mounting Bolting & Washers	COTS	0.6kg	£5.50
compressor_assembly-connector	High-Voltage IP65 Compressor Terminal Connector	COTS	0.15kg	£8.00
Module Total Cost:				£933.50

4.2.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The R290 inverter scroll compressor selection is thermodynamically highly efficient, but introduces severe A3 flammability and POE oil moisture-absorption risks to the assembly line. Corrective actions on acoustic materials and physical hardware protection for cold start liquid slugging are necessary before production.

Identified Issues

RISK

R290 is an A3 highly flammable refrigerant. Conventional assembly lines cannot charge or leak test A3 refrigerants without extreme explosion risk.

Suggestion: Require ATEX-certified explosion-proof zones for refrigerant charging, ventilation interlocks, spark-free tooling, and execute final leak checks using Helium mass spectrometry before R290 introduction.

MANUFACTURABILITY

Polyolester (POE) oil is aggressively hygroscopic. Open-to-air assembly time will allow moisture absorption, leading to acid formation and copper plating inside the circuit.

Suggestion: Limit atmospheric exposure of the compressor to less than 15 minutes. Mandate a deep dual-stage vacuum evacuation process (<500 microns) on the production line prior to charging.

MATERIALS

Mineral wool in the acoustic jacket is prone to absorbing atmospheric moisture and condensation from the cold surfaces of the compressor base, which will lead to shell corrosion over time.

Suggestion: Replace mineral wool with a closed-cell elastomeric acoustic foam (e.g., Nitrile Butadiene Rubber) combined with the mass-loaded vinyl.

ENGINEERING

Liquid slugging during cold starts is identified as a failure mode but no hardware mitigation is specified in the assembly module.

Suggestion: Incorporate a suitably sized suction accumulator and stringently specify a PTC crankcase heater interlocked with the control board to boil off liquid R290 prior to startup.

ENGINEERING

The mass of the compressor running at 120Hz can shear standard rubber mounts during shipping or high-torque startup events.

Suggestion: Ensure the vibration-isolating rubber feet utilize captive hardware (e.g., fail-safe mounts) so the compressor cannot rip free during transit or severe NVH events.

Strategic Recommendations

- Upgrade the final assembly and charging station to an ATEX-certified, explosion-proof environment with high-volume low-level exhaust ventilation.
- Implement strict time-tracking for compressor plug removal to ensure POE oil atmospheric exposure is under 15 minutes.
- BOM Change: Replace the mineral wool acoustic layer with moisture-repellent, closed-cell acoustic foam.
- BOM Change: Add a suction accumulator to the immediate inlet plumbing to physically decouple liquid slugging risks from the scroll involutes.
- BOM Change: Add a PTC crankcase heater wrapping the lower shell to maintain oil temperature limits during cold soaking.

4.3 Air-side Heat Exchanger (Evaporator)

Lead Time: 12 weeks

Status: draft

Module Purpose

Absorbs natural thermal energy from ambient air to vaporize liquid refrigerant.

Why it matters to the system

The sheer size and thermodynamic efficiency of the evaporator dictate the frost cycle frequency and the minimum air flow rate, directly impacting the final acoustic output and electrical fan load.

Detailed Technical Description

Constructed as a massive L-shaped or U-shaped array measuring approximately 1100mm (H) x 1800mm (L), this module provides the crucial 1.98 m² face area necessary to harvest ~24kW of ambient heat. It relies on 7mm internally grooved copper tubes traversing closely packed louvered aluminium fins (approx 2.1mm pitch). The design guarantees a low face velocity of 1.5 - 2.0 m/s for the required 12,600 m³/h volumetric airflow, effectively minimizing turbulent air noise and static pressure drop.

The fins are coated with a hydrophilic polymer designed to accelerate condensate drainage during periods of high humidity. This delays the inevitable frosting of the coil in the classic UK A2/W35 commercial envelope and facilitates rapid runoff during necessary reverse-cycle defrost operations. Intelligent distribution of the refrigerant via the header manifold guarantees an even phase change across the full face matrix, maintaining a close 6K-8K Delta T between ambient air and evaporating temperature.

SYSTEM INPUTS	SYSTEM OUTPUTS
- Ambient air at 12,600 m³/h - Low-pressure liquid/vapor R290 mix	- Cooled exhaust air - Low-pressure superheated R290 vapor - Condensate water

4.3.2 Specifications & Risk Profile

Expected Key Parts

- 7mm grooved copper tubing
- Hydrophilic aluminium louvered fins
- Multipass header manifold
- Defrost drip tray

Identified Failure Modes

- Uneven refrigerant distribution causing localized severe frosting and airflow blockage
- Galvanic corrosion between copper tubes and aluminium fins degrading thermal transfer
- Header pipe fatigue cracking from continuous thermal cycling (defrost expansions)

Open Questions

- The exact transient frost growth rate at 2°C and 90% humidity in the actual aerodynamic housing
- Optimal circuiting strategy to prevent refrigerant pooling at the bottom manifold runs

4.3.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
evaporator_coil-001	7mm grooved copper tubing	COTS	2.5kg	£40.00
evaporator_coil-002	Hydrophilic aluminium louvered fins	COTS	2.5kg	£40.00
evaporator_coil-003	Multipass header manifold	COTS	2.5kg	£40.00
evaporator_coil-004	Defrost drip tray	COTS	2.5kg	£40.00
evaporator_coil-brackets	Internal Structural Exoskeleton Brackets	Custom	3.5kg	£25.00
evaporator_coil-heater	Drip Tray Base Defrost Basepan Heater Cable	COTS	0.4kg	£18.00
Module Total Cost:				£203.00

4.3.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The evaporator design is thermally sound and employs industry-standard components, but bending a massive 1100x1800mm fin-and-tube assembly into an L or U shape presents significant manufacturability and reject-rate risks.

Identified Issues

MANUFACTURABILITY

Bending a large-scale fin-and-tube matrix into an L or U shape requires specialized CNC benders and strict process control to prevent fin crushing, tube buckling, or excessive wall thinning.

Suggestion: Conduct a tollgate review with the coil manufacturer to validate their bending equipment's capacity for this specific geometry and verify maximum fin pitch tolerance during bending.

ENGINEERING

Header pipe fatigue cracking from continuous thermal cycling (defrost expansions) on a large multi-pass assembly is a prevalent and costly warranty failure mode.

Suggestion: Introduce stress-relief loops (pigtails) or expansion bends between the header manifold branches and the main coil connections to absorb thermal strain.

MATERIALS

The copper/aluminum interface is a well-known site for galvanic corrosion, which can be exacerbated by hydrophilic coatings holding a moisture film.

Suggestion: Ensure the hydrophilic polymer is applied over a robust anti-corrosive epoxy primer (e.g., GoldFin/BlueFin equivalent) that completely coats the Cu/Al junctions.

RISK

Uneven refrigerant distribution across a massive 1.98 m² face will cause localized frosting, forcing premature and inefficient defrost cycles.

Suggestion: Utilize precision-machined distributor spiders and matched capillary tubes to ensure perfectly balanced liquid injection into each circuit.

Strategic Recommendations

- Validate assembly line CNC bending capability and specify acceptable tolerances for fin deformation at the bend radius.
- Implement thermal stress-relief geometries (bends/loops) on all rigid header/manifold connections.
- Specify an anti-corrosion e-coating process beneath the hydrophilic layer, particularly at the tube-to-fin collars.
- Require a low-temperature base pan heater combined with the active defrost drip tray to prevent ice build-up in A2/W35 UK climates.

4.4 EC Axial Fan Assembly

Lead Time: 10 weeks

Status: draft

Module Purpose

Induces necessary high-volume airflow across the evaporator coil to extract ambient thermal energy.

Why it matters to the system

The fan dictates whether the product legally passes ISO 3744 acoustic testing. Over-speeding fans are the singular cause of commercial heat pump noise complaints and failed planning permissions.

Detailed Technical Description

Crucial to solving the acoustic paradox of the brief, this assembly moves an immense 12,600 m³/h of air while aggressively suppressing acoustic output. A single large-diameter 710mm axial fan, rather than dual high-RPM 500mm fans, generates the required flow at an exceptionally low rotational speed (approx 750 RPM). Employing advanced Electronically Commutated (EC) DC motors and biometric or swept blade geometries (comparable to Ziehl-Abegg OWlet or ebm-papst AxiBlade), it yields a Sound Power Level below 62dB(A).

The assembly is structurally fixed to a precisely engineered cowl designed to mitigate tip-vortex shedding, one of the primary sources of broad-spectrum noise in outdoor heat pumps. The EC motor operates strictly on a variable basis, adjusting its RPM continually based on the exact thermal output requested by the control board and the dynamic static pressure imposed by progressive frost buildup on the evaporator coil.

SYSTEM INPUTS	SYSTEM OUTPUTS
230V AC Power - PWM continuous speed signal	12,600 m³/h airflow (approx 3.5 m³/s) - Aerodynamic noise profile

4.4.2 Specifications & Risk Profile

Expected Key Parts

- 710mm Swept/biometric blade impeller
- Electronically Commutated (EC) Motor
- Aerodynamic fan cowl/shroud
- Bird/finger guard grille

Identified Failure Modes

- Bearing seizure due to continual exposure to driving rain and freezing temperatures
- Blade imbalance propagating severe low-frequency vibration to the chassis
- EC motor control electronics failing under extreme outdoor thermal cycling

Open Questions

- True static pressure penalty incurred by the structural finger guard
- Exact acoustic directivity pattern of the fan relative to near-field ground reflections

4.4.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
axial_fan_assembly-001	710mm Swept/biometric blade impeller	COTS	2.5kg	£40.00
axial_fan_assembly-002	Electronically Commutated (EC) Motor	COTS	2.5kg	£40.00
axial_fan_assembly-003	Aerodynamic fan cowl/shroud	COTS	2.5kg	£200.00
axial_fan_assembly-004	Bird/finger guard grille	COTS	2.5kg	£40.00
axial_fan_assembly-spacers	Vibration Damping Spacers & Nut Kit	COTS	0.3kg	£6.50
axial_fan_assembly-harness	EC Motor Power & Modbus/PWM Wiring Harness	COTS	0.5kg	£14.00
Module Total Cost:				£340.50

4.4.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The strategy of using a single large-diameter, low-RPM fan is acoustically excellent, but the 710mm diameter introduces significant challenges regarding tip-clearance tolerances, structural stiffness, and ice-induced vibration.

Identified Issues

TOLERANCES

Maintaining tight tip-clearances to mitigate vortex shedding on a 710mm fan requires extreme structural stiffness. Wind loads or chassis flex could cause catastrophic blade strikes.

Suggestion: Engineer the cowl with heavy ribbing or deep-drawn flanges, and mount the motor via a rigid, triangulated bracket integrated directly into the strongest point of the chassis.

RISK

Ice buildup on large swept blades in freezing rain will cause massive imbalances and severe low-frequency (approx 12.5 Hz at 750 RPM) vibration.

Suggestion: Incorporate specialized elastomeric vibration isolators tuned precisely to the 12-15 Hz range, and implement a software de-icing routine (e.g., brief reverse rotation or periodic shedding spin).

MATERIALS

EC motor electronics are highly susceptible to outdoor thermal cycling and moisture ingress, leading to premature control failure.

Suggestion: Mandate an IP55 or IP68 rating for the fan assembly, ensuring all onboard EC electronics are fully potted with epoxy or polyurethane.

MANUFACTURABILITY

Tooling for a 710mm aerodynamic cowl is massive and expensive. If injection molded, it requires a very high-tonnage press.

Suggestion: Evaluate thermoforming for lower volumes, or utilize commercial-off-the-shelf (COTS) plug-and-play fan/cowl assemblies from suppliers like ebm-papst or Ziehl-Abegg to eliminate custom tooling costs.

Strategic Recommendations

- Source a pre-balanced COTS 710mm EC fan and cowl assembly (e.g., ebm-papst AxiBlade) rather than manufacturing a custom cowl, ensuring guaranteed aerodynamic and acoustic performance.
- Specify fully potted motor electronics and sealed, permanently lubricated bearings with a minimum IP55 weather rating.
- Integrate tuned elastomeric vibration isolators between the fan mounting bracket and the main structural chassis.
- Design a heated wire trace or thermal loop into the lower section of the cowl to prevent ice bridging between the blades and the shroud.

4.5 Water-side Heat Exchanger (Condenser)

Lead Time: 8 weeks

Status: draft

Module Purpose

Transfers heat from the hot, high-pressure refrigerant to the building's hydronic heating circuit.

Why it matters to the system

This is the primary constraint holding the A2/W35 baseline COP. Poor heat transfer here elevates required condensing temperature, immediately decaying thermodynamic efficiency.

Detailed Technical Description

This sub-assembly acts as the thermal bridge between the volatile R290 refrigerant cycle and the safe, enclosed water circuit. Using an asymmetric Brazed Plate Heat Exchanger (BPHE) constructed of 60 to 80 thin corrugated 316L stainless steel plates, it ensures highly turbulent flow on both the refrigerant and water paths. This turbulence maximizes the convective heat transfer coefficient, capable of transferring 30kW at standard W35 conditions with an exceedingly tight 3K to 5K Delta T approach.

The component is rigorously evaluated under the Pressure Equipment Directive (PED 2014/68/EU). Designed to handle the aggressive pressures of condensed propane (often exceeding 20 bar in high temp phases), the copper brazing and stainless matrix ensure zero cross-contamination. It also maintains a low water-side pressure drop (below 25 kPa), meaning standard commercial EC circulating pumps can easily overcome the resistance without imposing high auxiliary electrical loads.

SYSTEM INPUTS	SYSTEM OUTPUTS
- High-pressure hot R290 vapor - Return water from building (e.g., 30°C)	- High-pressure liquid R290 (subcooled) - Flow water to building (e.g., 35°C)

4.5.2 Specifications & Risk Profile

Expected Key Parts

- 316L Stainless steel stamped plates
- 99% pure copper brazing matrix
- Threaded water connection spigots
- Sweat-brazed refrigerant stubs

Identified Failure Modes

- Internal scaling/fouling from hard water side reducing thermal transfer coefficient
- Plate rupture due to water-side freezing during unpowered winter ambient drops
- Refrigerant channeling causing incomplete condensation and poor subcooling

Open Questions

- Risk of localized plate stressing during deep reverse-cycle defrost shocks
- Real-world fouling factor margin over long term deployment without chemical dosing

4.5.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
condenser_bphe-001	316L Stainless steel stamped plates	COTS	2.5kg	£40.00
condenser_bphe-002	99% pure copper brazing matrix	COTS	2.5kg	£40.00
condenser_bphe-003	Threaded water connection spigots	COTS	2.5kg	£40.00
condenser_bphe-004	Sweat-brazed refrigerant stubs	COTS	2.5kg	£40.00
condenser_bphe-insulation	BPHE Polyurethane Thermal Insulation Jacket	COTS	0.4kg	£22.00
condenser_bphe-bracket	Heavy-Duty BPHE Tie-Down Strap & Bracket	COTS	0.7kg	£9.00
Module Total Cost:				£191.00

4.5.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The use of an asymmetric 316L/copper BPHE is highly efficient and manufacturable, but the combination of high-pressure A3 flammable refrigerant (R290) and catastrophic rupture risks from water-side freezing necessitates stringent inline quality control and system-level fail-safes.

Identified Issues

RISK

Water-side freezing can rupture the plates, potentially causing high-pressure flammable R290 to inject directly into the building's enclosed hydronic water circuit.

Suggestion: Require an active anti-freeze pump routine in the control logic, automated drain valves, or mandate a glycol mixture for the hydronic circuit. Include a pressure relief valve on the water loop.

MANUFACTURABILITY

Sweat-brazing refrigerant stubs for R290 poses significant safety risks if joints are porous or fail under >20 bar cycling.

Suggestion: Implement 100% inline automated helium mass spectrometry leak testing for all brazed assemblies prior to refrigerant charging.

ENGINEERING

Internal scaling and fouling on the hydronic side will rapidly degrade the tight 3K-5K Delta T approach.

Suggestion: Mandate the installation of a Y-strainer and a magnetic sludge separator at the water inlet in the installation manual.

MATERIALS

99% pure copper brazing matrix is standard and robust, but susceptible to corrosion if the water chemistry (pH, chlorides) goes out of spec.

Suggestion: Add a specification for allowable water chemistry limits in the warranty documentation.

Strategic Recommendations

- Implement 100% inline helium leak testing for the entire refrigerant-side assembly to ensure PED and R290 safety compliance.
- Design and validate system-level freeze protection logic (e.g., running the circulating pump periodically when ambient temperatures drop below 3°C).
- Specify strict water quality requirements (max chloride levels, pH range) to prevent galvanic or pitting corrosion on the copper braze matrix.
- Ensure the stamping tool for the corrugated BPHE plates is monitored for wear to prevent refrigerant channeling and maintain the turbulent flow required for the 30kW transfer.

4.6 Refrigeration Control Train

Lead Time: 12 weeks

Status: draft

Module Purpose

Modulates refrigerant flow rates, handles phase expansion, and executes cycle reversals for defrosting.

Why it matters to the system

Without precise superheat management via the EEV, liquid refrigerant could return to the compressor (fatal slugging) or starve the evaporator (collapsing COP).

Detailed Technical Description

This complex network of brazed copper piping and intelligent valving controls the fundamental phase states of the R290 cycle. The primary actuator, the Electronic Expansion Valve (EEV), modulates a precise stepper motor to drop the pressure of the liquid propane into an atomized mist. Governed directly by the control PCB observing the evaporator exit superheat (targeting a strict 4-6K), the EEV adjusts dynamically relative to compressor speed, perfectly tracking the thermal load.

The inclusion of a 4-Way Reversing Valve is mandatory for the UK winter envelope, physically shifting the hot-gas discharge path to pump direct compressor heat backwards into the evaporator to shear structural frost. Crucially, the entirety of this circuit must be perfectly hermetic; under ATEX and DSEAR guidelines, no mechanical joints (like flares) are permissible—all junctions must be hard-brazed and factory helium leak-tested to prevent highly flammable A3 gas accumulation in the outdoor chassis.

SYSTEM INPUTS

- High-pressure liquid R290 (from condenser)
- Electronic PID logic signals
- Evaporator gas

SYSTEM OUTPUTS

- Low-pressure mist R290 (to evaporator)
- Reversed thermodynamic flow pathways

4.6.2 Specifications & Risk Profile

Expected Key Parts

- Electronic Expansion Valve (EEV)
- 4-Way reversing valve
- Suction line accumulator
- High/Low pressure transducers
- Dehydrator filter drier

Identified Failure Modes

- Stepper motor binding in the EEV, preventing system modulation
- 4-Way valve internal slide mechanism jamming during a freeze-defrost transition
- Micro-leak at a brazed joint causing slow loss of the 2.0kg R290 charge over time

Open Questions

- Transient response delay of the EEV during instant compressor inverter ramp-ups
- Acoustic hiss severity emanating from the EEV orifice under maximal pressure differential

4.6.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
refrigerant_valving_circuit-001	Electronic Expansion Valve (EEV)	COTS	2kg	£150.00
refrigerant_valving_circuit-002	4-Way reversing valve	COTS	2kg	£80.00
refrigerant_valving_circuit-003	Suction line accumulator	COTS	2kg	£40.00
refrigerant_valving_circuit-004	High/Low pressure transducers	COTS	2kg	£60.00
refrigerant_valving_circuit-005	Dehydrator filter drier	COTS	2kg	£40.00
refrigerant_valving_circuit-fittings	Copper Braze Fittings Set (Elbows, U-Bends, Tees)	COTS	1.5kg	£28.00
refrigerant_valving_circuit-clamps	Vibration Isolating P-Clips / Pipe Saddles	COTS	0.3kg	£5.00
Module Total Cost:				£403.00

4.6.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The hard-brazed, mechanical-joint-free approach is excellent for A3 refrigerant leak prevention, but the 2.0kg R290 charge requires stringent ATEX component sourcing and rigorous factory brazing controls to pass manufacturing audits.

Identified Issues

RISK

A 2.0kg charge of R290 (propane) poses extreme flammability risks. Any electrical components within the enclosure, including the EEV stator, pressure transducers, and 4-way valve solenoid, must be explicitly ATEX Zone 2 certified or intrinsically safe.

Suggestion: Require supplier DSEAR/ATEX certification for all electrical actuators and sensors in the train.

MANUFACTURABILITY

100% hard-brazing of complex copper assemblies greatly increases the risk of internal oxidation (cupric oxide scaling) if not perfectly controlled. This scale will swiftly clog the fine orifice of the EEV.

Suggestion: Mandate continuous nitrogen purging during all brazing operations and implement automated induction brazing stations for critical joints.

ENGINEERING

4-Way reversing valves are highly susceptible to internal slide jamming during freeze-defrost transitions, especially if subjected to oil logging or improper orientation.

Suggestion: Specify a strictly horizontal mounting orientation for the 4-Way valve on the assembly line and design the suction line piping to promote continuous oil return.

TOLERANCES

Maintaining a strict 4-6K superheat target leaves very little margin for sensor error or lag in the EEV response.

Suggestion: Pair the EEV with high-accuracy, fast-response pressure transducers (<1% full-scale error) and Class-A thermistors. Ensure the PCB control algorithm includes anti-hunting PID logic.

Strategic Recommendations

- Verify EX-ratings (ATEX Zone 2/Ex ic) for all electrical components situated in the refrigeration module enclosure.
- Introduce automated induction brazing with a verified nitrogen purge process to prevent inner oxidation and joint failure.
- Install a 100-mesh strainer immediately upstream of the EEV to intercept any brazing slag or debris.
- Set the factory helium leak test threshold to $< 1 \times 10^{-6}$ mbar L/s to guarantee absolute hermeticity for the R290 charge.

4.7 Sealed Electrical & Drive Enclosure

Lead Time: 16 weeks

Status: draft

Module Purpose

Provides safe, isolated power electronics, motor driving, and decision logic in a highly flammable environment.

Why it matters to the system

This module represents the critical intersection of functional efficiency (inverter algorithms) and explosive safety compliance (spark exclusion).

Detailed Technical Description

Due to IEC 60335-2-40 and BS EN 378 regulations covering flammable R290 refrigerants, traditional open electronics are strictly prohibited. The electrical control box is an IP65-rated, flame-proof micro-enclosure housed within the monobloc, isolating all capable ignition sources (contactors, high-voltage switching relays) from any potential refrigerant leak paths. Often placed upstream of the primary fan airflow or kept under a mild positive pressure, it ensures zero chance of a spark igniting a heavy propane cloud.

Functionally, it houses the sophisticated brain of the system: a purchased Tier-1 driver platform (e.g., Carel or Eliwell) encompassing the BLDC compressor inverter drive and the primary logic PCB. It interprets incoming ambient, coil, and water temperatures alongside system pressures to orchestrate fan speeds, compressor Hz, and EEV steps, executing the precise PID control required to push the system to an ErP A+++ Seasonal COP.

SYSTEM INPUTS	SYSTEM OUTPUTS
400V 3-phase or 230V 1-phase AC Power - Sensor thermistor data (NTCs) - Pressure transducer voltages	- High-voltage PWM (Compressor) - Low-voltage stepper signals (EEV) 0-10V Fan control - Diagnostic data logs

4.7.2 Specifications & Risk Profile

Expected Key Parts

- IP65-rated sealed metal enclosure
- Inverter drive / BLDC Controller
- Proprietary Heat Pump Main PCB
- ATEX mechanically isolated relays/contactors
- Wire harnesses

Identified Failure Modes

- Inverter IGBT thermal breakdown due to inadequate cooling of the sealed IP65 box
- Software logic loop leading to repeated short-cycling of the compressor
- Seal failure on the IP65 door allowing moisture ingress and PCB shorting

Open Questions

- Passive thermal dissipation capabilities of the sealed IP65 box under 30kW full-load inverter heat rejection
- Exact third-party firmware licensing constraints for bespoke defrost logic

4.7.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
electrical_control_box-001	IP65-rated sealed metal enclosure	Custom	2kg	£120.00
electrical_control_box-002	Inverter drive / BLDC Controller	COTS	2kg	£400.00
electrical_control_box-003	Proprietary Heat Pump Main PCB	COTS	2kg	£250.00
electrical_control_box-004	ATEX mechanically isolated relays/contactors	COTS	2kg	£40.00
electrical_control_box-005	Wire harnesses	COTS	2kg	£45.00
electrical_control_box-glands	IP65 Cable Glands (M20/M25/M32 Set)	COTS	0.2kg	£8.50
electrical_control_box-terminals	DIN Rail & Terminal Block Assembly	COTS	0.6kg	£16.00
Module Total Cost:				£879.50

4.7.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The R290 safety compliance strategy is fundamentally sound, but housing a high-power inverter inside a completely sealed IP65 enclosure presents a severe thermal management risk that needs immediate mechanical resolution.

Identified Issues

ENGINEERING

IGBT thermal breakdown is almost guaranteed if the inverter drive is fully enclosed in a sealed IP65 metal box without a dedicated thermal exhaust path.

Suggestion: Redesign the enclosure backplate to allow the inverter's heatsink to pass through to the outside, exposing it to the primary fan airflow while maintaining a perimeter IP65 seal.

MANUFACTURABILITY

Sealing numerous wire harnesses (power, sensors, comms) using individual cable glands is highly labor-intensive and introduces multiple points for operator error, leading to moisture ingress.

Suggestion: Switch to a molded elastomeric multi-cable transit frame or a pre-potted bulkhead connector to error-proof wire harness passthroughs.

RISK

Maintaining 'mild positive pressure' adds significant pneumatic complexity, requiring pumps, tubing, and pressure sensors that could fail over a 15-year lifecycle.

Suggestion: If the box is truly IP65 and utilizes ATEX-rated contactors, consider removing the active positive pressure system to reduce BOM cost and eliminate a failure mode.

MATERIALS

The IP65 door seal material must withstand broad temperature fluctuations and ozone exposure without taking a compression set that would compromise the seal over time.

Suggestion: Specify high-grade, closed-cell Silicone or EPDM gaskets with a documented compression set resistance over 10+ years.

Strategic Recommendations

- Implement a 'through-wall' heatsink design for the inverter drive to decouple internal electronic sealing from external thermal management.
- Adopt multi-hole, quick-install cable transit glands to streamline assembly and guarantee IP65 ratings on the factory floor.
- Strip out the active positive pressure subsystem to reduce parts count, assuming the IP65 rating and ATEX components satisfy the rigid IEC 60335-2-40 Annex LL requirements.
- Ensure dedicated physical separation (via molded plastic barriers inside the enclosure) between high-voltage inverter lines and intrinsically safe low-voltage sensor inputs.

4.8 Indoor Hydronic Control Station

Lead Time: 8 weeks

Status: draft

Module Purpose

Circulates the heated water throughout the commercial building while keeping all flammable refrigerant strictly outdoors.

Why it matters to the system

This module alone solves the BS EN 378 compliance hurdle, completely bypassing the complex room-volume capacity calculations that kill R290 indoor deployments, ensuring standard commercial deployability.

Detailed Technical Description

The Indoor Hydronic Box represents the fundamental split-system architectural pivot required to bypass severe indoor A3 flammability limits. By isolating the thermodynamic cycle entirely in the outdoor monobloc, zero propane enters the building footprint. This wall-hung indoor unit focuses solely on hydronic distribution, encompassing the circulation pumps, expansion vessel, valving, safety blowoffs, and an integrated backup electric immersion heater to supplement peak winter loads.

It additionally serves as the primary system interface for facility managers and installers. Housing a sophisticated Human-Machine Interface (HMI) panel, it bridges the gap between the outdoor thermal production and indoor heat demand. It handles real-time flow rate sensing and communicates via standard commercial BMS protocols (such as Modbus or BACnet), mapping directly to Microgeneration Certification Scheme (MCS) requirements for quantifiable heat metering.

SYSTEM INPUTS	SYSTEM OUTPUTS
- Heated flow water from outdoor monobloc - Return water from commercial building heating loop - User interface requests	- Pressurized heated water delivery - System diagnostics mapping

4.8.2 Specifications & Risk Profile

Expected Key Parts

- High-efficiency EC Circulation Pump (e.g., Wilo Stratos MAXO)
- Diaphragm expansion vessel
- Backup electric immersion heater
- Flow and return temperature sensors
- Smart HMI Touchscreen

Identified Failure Modes

- Pump cavitation due to inadequate system pressure or micro-bubble accumulation
- Backup immersion contactor fusing, remaining permanently energized
- Loss of communication protocol integrity with outdoor monobloc control board

Open Questions

- Variability of pre-existing commercial pipework resistances impacting pump curve selection
- Exact expansion vessel sizing capable of covering wildly varying legacy buffer tank volumes

4.8.3 Cost Breakdown

PART #	NAME	TYPE	MASS	COST
indoor_hydronic_box-001	High-efficiency EC Circulation Pump (e.g., Wilo Stratos MAXO)	COTS	2kg	£250.00
indoor_hydronic_box-002	Diaphragm expansion vessel	COTS	2kg	£40.00
indoor_hydronic_box-003	Backup electric immersion heater	COTS	2kg	£40.00
indoor_hydronic_box-004	Flow and return temperature sensors	COTS	2kg	£60.00
indoor_hydronic_box-005	Smart HMI Touchscreen	COTS	2kg	£180.00
indoor_hydronic_box-bracket	Heavy-Duty Wall Mounting Bracket Cleat	Custom	2.2kg	£15.00
indoor_hydronic_box-safety	3-Bar Pressure Relief Valve & Auto Air Vent Set	COTS	0.8kg	£25.00
Module Total Cost:				£610.00

4.8.4 Engineering Review

Review by Fang

WARN

Verdict Summary

The system architecture successfully mitigates A3 refrigerant flammability risks by isolating it outdoors, which is excellent. However, the failure modes related to pump cavitation and thermal runaway of the backup heater require immediate mechanical and electrical safeguards before production.

Identified Issues

RISK

Backup electric immersion contactor fusing can lead to uncontrolled heating, potentially boiling the hydronic loop and causing catastrophic pressure buildup.

Suggestion: Implement an independent, manual-reset electromechanical high-limit thermal cutout switch directly in series with the heater power supply.

ENGINEERING

Pump cavitation due to micro-bubble accumulation decreases efficiency and will destroy the EC circulation pump over time.

Suggestion: Add a micro-bubble air and dirt separator (e.g., Spirovent) at the hottest point of the circuit prior to pump suction, along with an auto air vent (AAV).

MANUFACTURABILITY

Integrating a heavy Wilo Stratos MAXO pump, a water-filled expansion vessel, and copper/brass manifolds into a wall-hung unit introduces high dynamic loads and vibration transmission.

Suggestion: Utilize a reinforced structural backplate with elastomeric mounting grommets for the pump to minimize resonant wall vibration and simplify assembly.

RISK

Loss of communication protocol with the outdoor monobloc could leave the system in an undetermined state during active heating calls.

Suggestion: Establish a failsafe hardware heartbeat; upon comms loss, the pump should enter a default circulation mode to prevent freezing, and trigger a local HMI alarm.

Strategic Recommendations

- Incorporate a hardwired, independent thermal safety limit switch for the backup immersion heater.
- Design an internal piping manifold that includes a dedicated micro-bubble separator and dirt filter before the pump inlet.
- Perform vibration and load testing on the wall-bracket chassis to ensure the sheet metal can support the fully flooded weight of the components.
- Use pre-fabricated pipe assemblies with O-ring face seals or press-fittings within the enclosure to speed up manufacturing assembly and reduce brazing inside the cabinet constraints.

5. BILL OF MATERIALS (BOM)

5.1 BOM — Structural & Acoustic Enclosure

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
structural_enclosure-001	Powder-coated Z275 galvanized steel panels	purchased_cots / varies	COTS	2kg	£40.00
structural_enclosure-002	Welded skeletal chassis	purchased_cots / varies	COTS	2kg	£120.00
structural_enclosure-003	Mass-loaded acoustic foam baffles	purchased_cots / varies	COTS	2kg	£40.00
structural_enclosure-004	Rubber-in-shear chassis mounts	purchased_cots / varies	COTS	2kg	£120.00
structural_enclosure-005	ATEX ventilation grilles	purchased_cots / varies	COTS	2kg	£40.00
structural_enclosure-fasteners	M8 Stainless Steel Assembly Fasteners Kit	purchased_cots / Stainless Steel 304	COTS	1.2kg	£15.00
structural_enclosure-gaskets	EPDM Weatherproof Panel Gasketing	purchased_cots / EPDM Rubber	COTS	0.8kg	£12.50
structural_enclosure-labels	ATEX / Flammable Gas Warning Labels	purchased_cots / Vinyl	COTS	0.05kg	£3.00

5. BILL OF MATERIALS (BOM)

5.2 BOM — Inverter Scroll Compressor Assembly

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
compressor_assembly-001	BLDC R290 Scroll Compressor	purchased_cots / varies	COTS	2.5kg	£800.00
compressor_assembly-002	Polyolester (POE) lubricating oil	purchased_cots / varies	COTS	2.5kg	£40.00
compressor_assembly-003	Mass-loaded vinyl acoustic jacket	purchased_cots / varies	COTS	2.5kg	£40.00
compressor_assembly-004	Vibration-isolating rubber feet	purchased_cots / varies	COTS	2.5kg	£40.00
compressor_assembly-hardware	M10 Compressor Mounting Bolting & Washers	purchased_cots / Galvanized Steel	COTS	0.6kg	£5.50
compressor_assembly-connector	High-Voltage IP65 Compressor Terminal Connector	purchased_cots / High-Temp Polymer / Brass	COTS	0.15kg	£8.00

5. BILL OF MATERIALS (BOM)

5.3 BOM — Air-side Heat Exchanger (Evaporator)

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
evapora- tor_coil-001	7mm grooved copper tubing	purchased_cots / varies	COTS	2.5kg	£40.00
evapora- tor_coil-002	Hydrophilic aluminium louvered fins	purchased_cots / varies	COTS	2.5kg	£40.00
evapora- tor_coil-003	Multipass header manifold	purchased_cots / varies	COTS	2.5kg	£40.00
evapora- tor_coil-004	Defrost drip tray	purchased_cots / varies	COTS	2.5kg	£40.00
evapora- tor_coil-brackets	Internal Structural Exoskeleton Brackets	sheet_metal_bending / Galvanized Steel	Custom	3.5kg	£25.00
evapora- tor_coil-heater	Drip Tray Base Defrost Basepan Heater Cable	purchased_cots / Silicone Insulation / Copper	COTS	0.4kg	£18.00

5. BILL OF MATERIALS (BOM)

5.4 BOM — EC Axial Fan Assembly

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
axial_fan_assembly-001	710mm Swept/biometric blade impeller	purchased_cots / varies	COTS	2.5kg	£40.00
axial_fan_assembly-002	Electronically Commutated (EC) Motor	purchased_cots / varies	COTS	2.5kg	£40.00
axial_fan_assembly-003	Aerodynamic fan cowl/shroud	purchased_cots / varies	COTS	2.5kg	£200.00
axial_fan_assembly-004	Bird/finger guard grille	purchased_cots / varies	COTS	2.5kg	£40.00
axial_fan_assembly-spacers	Vibration Damping Spacers & Nut Kit	purchased_cots / Rubber / Steel	COTS	0.3kg	£6.50
axial_fan_assembly-harness	EC Motor Power & Mod-bus/PWM Wiring Harness	purchased_cots / Copper / PVC	COTS	0.5kg	£14.00

5. BILL OF MATERIALS (BOM)

5.5 BOM — Water-side Heat Exchanger (Condenser)

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
con-denser_bphe-001	316L Stainless steel stamped plates	purchased_cots / varies	COTS	2.5kg	£40.00
con-denser_bphe-002	99% pure copper brazing matrix	purchased_cots / varies	COTS	2.5kg	£40.00
con-denser_bphe-003	Threaded water connection spigots	purchased_cots / varies	COTS	2.5kg	£40.00
con-denser_bphe-004	Sweat-brazed refrigerant stubs	purchased_cots / varies	COTS	2.5kg	£40.00
con-denser_bphe-insulation	BPHE Polyurethane Thermal Insulation Jacket	purchased_cots / Polyurethane Foam	COTS	0.4kg	£22.00
con-denser_bphe-bracket	Heavy-Duty BPHE Tie-Down Strap & Bracket	purchased_cots / Stainless Steel 304	COTS	0.7kg	£9.00

5. BILL OF MATERIALS (BOM)

5.6 BOM — Refrigeration Control Train

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
refrigerant_valving_circuit-001	Electronic Expansion Valve (EEV)	purchased_cots / varies	COTS	2kg	£150.00
refrigerant_valving_circuit-002	4-Way reversing valve	purchased_cots / varies	COTS	2kg	£80.00
refrigerant_valving_circuit-003	Suction line accumulator	purchased_cots / varies	COTS	2kg	£40.00
refrigerant_valving_circuit-004	High/Low pressure transducers	purchased_cots / varies	COTS	2kg	£60.00
refrigerant_valving_circuit-005	Dehydrator filter drier	purchased_cots / varies	COTS	2kg	£40.00
refrigerant_valving_circuit-fittings	Copper Braze Fittings Set (Elbows, U-Bends, Tees)	purchased_cots / Copper	COTS	1.5kg	£28.00
refrigerant_valving_circuit-clamps	Vibration Isolating P-Clips / Pipe Saddles	purchased_cots / Steel / EPDM	COTS	0.3kg	£5.00

5. BILL OF MATERIALS (BOM)

5.7 BOM — Sealed Electrical & Drive Enclosure

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
electrical_control_box-001	IP65-rated sealed metal enclosure	cnc / varies	Custom	2kg	£120.00
electrical_control_box-002	Inverter drive / BLDC Controller	purchased_cots / varies	COTS	2kg	£400.00
electrical_control_box-003	Proprietary Heat Pump Main PCB	purchased_cots / FR4	COTS	2kg	£250.00
electrical_control_box-004	ATEX mechanically isolated relays/contactors	purchased_cots / varies	COTS	2kg	£40.00
electrical_control_box-005	Wire harnesses	purchased_cots / varies	COTS	2kg	£45.00
electrical_control_box-glands	IP65 Cable Glands (M20/M25/M32 Set)	purchased_cots / Nylon	COTS	0.2kg	£8.50
electrical_control_box-terminals	DIN Rail & Terminal Block Assembly	purchased_cots / Mixed Plastics / Metal	COTS	0.6kg	£16.00

5. BILL OF MATERIALS (BOM)

5.8 BOM — Indoor Hydronic Control Station

PART #	NAME	MATERIAL/PROCESS	TYPE	MASS	COST
indoor_hydr-on-ic_box-001	High-efficiency EC Circulation Pump (e.g., Wilo Stratos MAXO)	purchased_cots / varies	COTS	2kg	£250.00
indoor_hydr-on-ic_box-002	Diaphragm expansion vessel	purchased_cots / varies	COTS	2kg	£40.00
indoor_hydr-on-ic_box-003	Backup electric immersion heater	purchased_cots / varies	COTS	2kg	£40.00
indoor_hydr-on-ic_box-004	Flow and return temperature sensors	purchased_cots / varies	COTS	2kg	£60.00
indoor_hydr-on-ic_box-005	Smart HMI Touchscreen	purchased_cots / varies	COTS	2kg	£180.00
indoor_hydr-on-ic_box-bracket	Heavy-Duty Wall Mounting Bracket Cleat	sheet_metal_bending / Powder-Coated Steel	Custom	2.2kg	£15.00
indoor_hydr-on-ic_box-safety	3-Bar Pressure Relief Valve & Auto Air Vent Set	purchased_cots / Brass	COTS	0.8kg	£25.00

6. Cost Waterfall & Economics

UNIT COST	TARGET CEILING	HEADROOM
£5,926.50	None	N/A

Make vs Buy Split

COTS (BUY) TOTAL	CUSTOM (MAKE) TOTAL	MAKE %
£3,791.00	£160.00	2.7%

Per-Module Roll-up

MODULE	% OF UNIT	COST
Module structural_enclosure	9.9%	£585.75
Module compressor_assembly	23.6%	£1,400.25
Module evaporator_coil	5.1%	£304.50
Module axial_fan_assembly	8.6%	£510.75
Module condenser_bphe	4.8%	£286.50
Module refrigerant_valving_circuit	10.2%	£604.50
Module electrical_control_box	22.3%	£1,319.25
Module indoor_hydraulic_box	15.4%	£915.00

NRE & Overheads

Total NRE (Tooling, Certs)	£4,800.00
Complexity Overhead Multiplier	1.5×

Grand Total (Unit + NRE)	£10,726.50
--------------------------	------------

Cost Confidence Assessment

Costing methodology employs deterministic expansions over a historical parts library combined with spot-price indices for raw materials. Custom parts are calculated volumetrically and adjusted by domain-specific complexity multipliers to ensure realistic unit economics.

Make-versus-buy categorisation is automated based on process availability. Non-recurring engineering (NRE) totals reflect tooling, testing, and compliance costs amortised over the target batch size. Margins of error apply to COTS components subject to global supply chain volatility.

7. RISKS REGISTER (FMEA)

7.1 Structural & Acoustic Enclosure Risks

Risk 1: Harmonic resonance amplification

Base Score: S:4 × L:3 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Perform finite element analysis (FEA) on chassis to shift natural frequency away from 12.5Hz (750 RPM).

Owner: Mechanical Engineer

Risk 2: Refrigerant pooling in chassis base

Base Score: S:5 × L:2 = 10

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Design sloped, unsealed floor pans allowing heavier-than-air R290 to vent downward safely.

Owner: Safety / DSEAR Officer

7. RISKS REGISTER (FMEA)

7.1 Structural & Acoustic Enclosure Risks

Risk 3: Acoustic foam degradation

Base Score: S:3 × L:3 = 9

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Specify closed-cell, non-absorbent foam to resist condensation and freezing cycles.
Owner: NVH Engineer	

7. RISKS REGISTER (FMEA)

7.2 Inverter Scroll Compressor Assembly Risks

Risk 4: Liquid slugging upon startup

Base Score: S:5 × L:3 = 15

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Integrate software-limited soft start and physical suction accumulator in the refrigeration loop.

Owner: Thermodynamics Engineer

Risk 5: Oil wash-out from high R290 solubility

Base Score: S:4 × L:2 = 8

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Limit discharge superheat and perform rigorous oil-return cycle tests at minimum Hz.

Owner: Thermodynamics Engineer

7. RISKS REGISTER (FMEA)

7.2 Inverter Scroll Compressor Assembly Risks

Risk 6: Acoustic flanking paths through pipework

Base Score: S:3 × L:4 = 12

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Install vibration-absorbing flexible piping loops directly at the compressor discharge and suction ports.

Owner: NVH Engineer

7. RISKS REGISTER (FMEA)

7.3 Air-side Heat Exchanger (Evaporator) Risks

Risk 7: Rapid frost accumulation blocking coil

Base Score: S:4 × L:4 = 16

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Implement predictive defrost logic utilizing ambient delta-T and fan current drops rather than generic timers.
Owner: Control Systems Engineer	

Risk 8: Refrigerant maldistribution

Base Score: S:3 × L:3 = 9

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Utilize CFD combined with an engineered distributor head to ensure equal two-phase mix to all vertical circuits.
Owner: Thermodynamics Engineer	

7. RISKS REGISTER (FMEA)

7.3 Air-side Heat Exchanger (Evaporator) Risks

Risk 9: Coil structural damage in transit

Base Score: S:2 × L:2 = 4

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Design bespoke rigid internal structural stays acting as exoskeleton for the L-coil.

Owner: Packaging Engineer

7. RISKS REGISTER (FMEA)

7.4 EC Axial Fan Assembly Risks

Risk 10: Exceeding 65dB(A) limit at full load

Base Score: S:5 × L:3 = 15

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Extensive wind-tunnel and anechoic chamber prototype testing; cap maximum PWM signal algorithmically.

Owner: Acoustics/NVH Engineer

Risk 11: Ice build-up on fan blades causing imbalance

Base Score: S:4 × L:2 = 8

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Design sufficient coil clearance; utilize blade materials inherent with hydrophobic properties.

Owner: Mechanical Engineer

7. RISKS REGISTER (FMEA)

7.4 EC Axial Fan Assembly Risks

Risk 12: Stall condition under heavy frost

Base Score: S:3 × L:3 = 9

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Continuously monitor motor current to proactively initiate defrost before pressure exceeds fan curve capabilities.
Owner: Control Systems Engineer	

7. RISKS REGISTER (FMEA)

7.5 Water-side Heat Exchanger (Condenser) Risks

Risk 13: Freezing rupture of water side

Base Score: S:5 × L:3 = 15

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Mandate glycol dosing in commissioning manual or incorporate automated pump cycling in sub-zero ambient conditions.
Owner: Product Manager	

Risk 14: Internal brazing failure under PED testing

Base Score: S:4 × L:1 = 4

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Source from Tier-1 EU manufacturers (e.g. Alfa Laval) holding full PED module H certifications.
Owner: Manufacturing Engineer	

7. RISKS REGISTER (FMEA)

7.5 Water-side Heat Exchanger (Condenser) Risks

Risk 15: High pressure drop eroding pump efficiency

Base Score: S:2 × L:2 = 4

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Specify asymmetric plate design maximizing water channel width over refrigerant channel width.

Owner: Thermodynamics Engineer

7. RISKS REGISTER (FMEA)

7.6 Refrigeration Control Train Risks

Risk 16: R290 slow leak at brazed joints

Base Score: S:5 × L:2 = 10

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Enforce strict EN 13134 brazing procedures and automated helium mass spectrometer leak testing at end-of-line.

Owner: Manufacturing Engineer

Risk 17: EEV modulation hunting

Base Score: S:3 × L:4 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Iteratively tune PID loops in firmware to decouple EEV response from compressor inverter ramps.

Owner: Control Systems Engineer

7. RISKS REGISTER (FMEA)

7.6 Refrigeration Control Train Risks

Risk 18: Liquid return during defrost

Base Score: S:4 × L:3 = 12

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Install an appropriately sized suction accumulator in the train to trap liquid pulses before the compressor inlet.
Owner: Thermodynamics Engineer	

7. RISKS REGISTER (FMEA)

7.7 Sealed Electrical & Drive Enclosure Risks

Risk 19: Inverter thermal overload

Base Score: S:4 × L:4 = 16

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Integrate a passive cold plate bonded to the IP65 enclosure back wall, cooled directly by the suction line or fan stream.
Owner: Thermal Engineer	

Risk 20: Spark ignition of pooled R290

Base Score: S:5 × L:1 = 5

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Mount enclosure entirely at the top of the chassis (R290 is heavier than air) and exclusively use solid-state or ATEX-rated relays.
Owner: Safety / DSEAR Officer	

7. RISKS REGISTER (FMEA)

7.7 Sealed Electrical & Drive Enclosure Risks

Risk 21: Supplier semiconductor shortages

Base Score: S:3 × L:3 = 9

CAUSE	CONSEQUENCE
Unknown	Unknown
EXISTING CONTROLS	PLANNED MITIGATION
None	Design PCB carrier board to be agnostic/compatible with at least two major drive suppliers (Carel and Eliwell).
Owner: Supply Chain Manager	

7. RISKS REGISTER (FMEA)

7.8 Indoor Hydronic Control Station Risks

Risk 22: Pump dead-heading against closed building valves

Base Score: S:4 × L:3 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Integrate an internal mechanical bypass valve and utilize a pump with integrated auto-variable pressure control.

Owner: Mechanical Engineer

Risk 23: Wiring fault isolating outdoor and indoor units

Base Score: S:3 × L:4 = 12

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Utilize robust, twisted-pair shielded RS485 communication lines specifically immune to EMI.

Owner: Electrical Engineer

7. RISKS REGISTER (FMEA)

7.8 Indoor Hydronic Control Station Risks

Risk 24: Customer resistance to split-hydronic footprint

Base Score: S:2 × L:2 = 4

CAUSE

Unknown

CONSEQUENCE

Unknown

EXISTING CONTROLS

None

PLANNED MITIGATION

Focus on high-aesthetic industrial design for the wallbox to make it feel analogous to a high-end commercial boiler interface.

Owner: Product Manager

8. SUPPLIER SHORTLIST & SOURCING

8.1 Powder-coated Z275 galvanized steel panels (structural_enclosure-001)

8. SUPPLIER SHORTLIST & SOURCING

8.2 Welded skeletal chassis (structural_enclosure-002)

8. SUPPLIER SHORTLIST & SOURCING

8.3 Mass-loaded acoustic foam baffles (structural_enclosure-003)

8. SUPPLIER SHORTLIST & SOURCING

8.4 Rubber-in-shear chassis mounts (structural_enclosure-004)

8. SUPPLIER SHORTLIST & SOURCING

8.5 ATEX ventilation grilles (structural_enclosure-005)

8. SUPPLIER SHORTLIST & SOURCING

8.6 BLDC R290 Scroll Compressor (compressor_assembly-001)

8. SUPPLIER SHORTLIST & SOURCING

8.7 Polyolester (POE) lubricating oil (compressor_assembly-002)

8. SUPPLIER SHORTLIST & SOURCING

8.8 Mass-loaded vinyl acoustic jacket (compressor_assembly-003)

8. SUPPLIER SHORTLIST & SOURCING

8.9 Vibration-isolating rubber feet (compressor_assembly-004)

8. SUPPLIER SHORTLIST & SOURCING

8.10 7mm grooved copper tubing (evaporator_coil-001)

8. SUPPLIER SHORTLIST & SOURCING

8.11 Hydrophilic aluminium louvered fins (evaporator_coil-002)

8. SUPPLIER SHORTLIST & SOURCING

8.12 Multipass header manifold (evaporator_coil-003)

8. SUPPLIER SHORTLIST & SOURCING

8.13 Defrost drip tray (evaporator_coil-004)

8. SUPPLIER SHORTLIST & SOURCING

8.14 710mm Swept/biometric blade impeller (axial_fan_assembly-001)

8. SUPPLIER SHORTLIST & SOURCING

8.15 Electronically Commutated (EC) Motor (axial_fan_assembly-002)

8. SUPPLIER SHORTLIST & SOURCING

8.16 Aerodynamic fan cowl/shroud (axial_fan_assembly-003)

8. SUPPLIER SHORTLIST & SOURCING

8.17 Bird/finger guard grille (axial_fan_assembly-004)

8. SUPPLIER SHORTLIST & SOURCING

8.18 316L Stainless steel stamped plates (condenser_bphe-001)

8. SUPPLIER SHORTLIST & SOURCING

8.19 99% pure copper brazing matrix (condenser_bphe-002)

8. SUPPLIER SHORTLIST & SOURCING

8.20 Threaded water connection spigots (condenser_bphe-003)

8. SUPPLIER SHORTLIST & SOURCING

8.21 Sweat-brazed refrigerant stubs (condenser_bphe-004)

8. SUPPLIER SHORTLIST & SOURCING

8.22 Electronic Expansion Valve (EEV) (refrigerant_valving_circuit-001)

8. SUPPLIER SHORTLIST & SOURCING

8.23 4-Way reversing valve (refrigerant_valving_circuit-002)

8. SUPPLIER SHORTLIST & SOURCING

8.24 Suction line accumulator (refrigerant_valving_circuit-003)

8. SUPPLIER SHORTLIST & SOURCING

8.25 High/Low pressure transducers (refrigerant_valving_circuit-004)

8. SUPPLIER SHORTLIST & SOURCING

8.26 Dehydrator filter drier (refrigerant_valving_circuit-005)

8. SUPPLIER SHORTLIST & SOURCING

8.27 IP65-rated sealed metal enclosure (electrical_control_box-001)

8. SUPPLIER SHORTLIST & SOURCING

8.28 Inverter drive / BLDC Controller (electrical_control_box-002)

8. SUPPLIER SHORTLIST & SOURCING

8.29 Proprietary Heat Pump Main PCB (electrical_control_box-003)

8. SUPPLIER SHORTLIST & SOURCING

8.30 ATEX mechanically isolated relays/contactors (electrical_control_box-004)

8. SUPPLIER SHORTLIST & SOURCING

8.31 Wire harnesses (electrical_control_box-005)

8. SUPPLIER SHORTLIST & SOURCING

8.32 High-efficiency EC Circulation Pump (e.g., Wilo Stratos MAXO) (indoor_hydronic_box-001)

8. SUPPLIER SHORTLIST & SOURCING

8.33 Diaphragm expansion vessel (indoor_hydronic_box-002)

8. SUPPLIER SHORTLIST & SOURCING

8.34 Backup electric immersion heater (indoor_hydronic_box-003)

8. SUPPLIER SHORTLIST & SOURCING

8.35 Flow and return temperature sensors (indoor_hydronic_box-004)

8. SUPPLIER SHORTLIST & SOURCING

8.36 Smart HMI Touchscreen (indoor_hydronic_box-005)

8. SUPPLIER SHORTLIST & SOURCING

8.37 M8 Stainless Steel Assembly Fasteners Kit (structural_enclosure-fasteners)

8. SUPPLIER SHORTLIST & SOURCING

8.38 EPDM Weatherproof Panel Gasketing (structural_enclosure-gaskets)

8. SUPPLIER SHORTLIST & SOURCING

8.39 ATEX / Flammable Gas Warning Labels (structural_enclosure-labels)

8. SUPPLIER SHORTLIST & SOURCING

8.40 M10 Compressor Mounting Bolting & Washers (compressor_assembly-hardware)

8. SUPPLIER SHORTLIST & SOURCING

8.41 High-Voltage IP65 Compressor Terminal Connector (compressor_assembly-connector)

8. SUPPLIER SHORTLIST & SOURCING

8.42 Internal Structural Exoskeleton Brackets (evaporator_coil-brackets)

8. SUPPLIER SHORTLIST & SOURCING

8.43 Drip Tray Base Defrost Basepan Heater Cable (evaporator_coil-heater)

8. SUPPLIER SHORTLIST & SOURCING

8.44 Vibration Damping Spacers & Nut Kit (axial_fan_assembly-spacers)

8. SUPPLIER SHORTLIST & SOURCING

8.45 EC Motor Power & Modbus/PWM Wiring Harness (axial_fan_assembly-harness)

8. SUPPLIER SHORTLIST & SOURCING

8.46 BPHE Polyurethane Thermal Insulation Jacket (condenser_bphe-insulation)

8. SUPPLIER SHORTLIST & SOURCING

8.47 Heavy-Duty BPHE Tie-Down Strap & Bracket (condenser_bphe-bracket)

8. SUPPLIER SHORTLIST & SOURCING

8.48 Copper Braze Fittings Set (Elbows, U-Bends, Tees) (refrigerant_valving_circuit-fittings)

8. SUPPLIER SHORTLIST & SOURCING

8.49 Vibration Isolating P-Clips / Pipe Saddles (refrigerant_valving_circuit-clamps)

8. SUPPLIER SHORTLIST & SOURCING

8.50 IP65 Cable Glands (M20/M25/M32 Set) (electrical_control_box-glands)

8. SUPPLIER SHORTLIST & SOURCING

8.51 DIN Rail & Terminal Block Assembly (electrical_control_box-terminals)

8. SUPPLIER SHORTLIST & SOURCING

8.52 Heavy-Duty Wall Mounting Bracket Cleat (indoor_hydronic_box-bracket)

8. SUPPLIER SHORTLIST & SOURCING

8.53 3-Bar Pressure Relief Valve & Auto Air Vent Set (indoor_hydronic_box-safety)

9. Audit Log

Pipeline execution log — generated by PDF Engine v2.1.

This document was generated automatically. Contents represent the synthesised output of the specialist pipeline.

Pipeline Execution Summary

Project Identifier	local-1777884295150
Research Result	Generated
System Modules	8 Modules
BOM Lines	53 Parts
Cost Analysis	Complete
Specialist Reviews	8 Completed
Supplier Shortlists	53 Matched

10. Source Attribution

This table documents the origin of data for each major section of the report, ensuring traceability of LLM outputs versus deterministic or user-provided data.

SECTION	SOURCE	PROVIDER / SYSTEM	DETAIL
Research	LLM	OpenRouter	Gemini 3.1 Pro via OpenRouter
Research	USER	user	Founder brief text
Regulatory	LLM	Unknown LLM	Gemini 3.1 Pro — standards extraction
Modules	LLM	Unknown LLM	Gemini 3.1 Pro — module decomposition
BOM	DETERMINISTIC	deterministic	Deterministic expansion from Max keyParts
BOM	LLM	Unknown LLM	Gemini 3.1 Pro — gap-fill for missing standard hardware
Cost	DETERMINISTIC	deterministic	Domain overhead multiplier model
Suppliers	SEARCH	search	Brave Search API — commercial supplier matching
Reviews	LLM	Unknown LLM	Gemini 3.1 Pro — Fang engineering review
Proofreader	LLM	Unknown LLM	Gemini 3.1 Pro — cross-module consistency check

11. Quality Scores

Automated self-critique mechanism evaluating the completeness, realism, and engineering validity of each section.

Brief

Score: 7/10

EVALUATION REASONS

- Use case is detailed
- Compliance notes present

IMPROVEMENT SUGGESTIONS

- Add mission statement
- Define target customers
- Add why-now rationale
- Declare cost ceiling
- Declare mass constraint

Regulatory

Score: 8/10

EVALUATION REASONS

- 8 standards identified
- Comprehensive regulatory coverage
- Most standards have applicability detail
- Most standards have design impact analysis
- Gap actions defined for some standards

Sizing

Score: 6/10

EVALUATION REASONS

- Design is infeasible — shown as exception
- Module dimensions computed
- 1 conflicts identified
- Recommendations provided

Modules

Score: 9/10

EVALUATION REASONS

- 8 modules — good granularity
- Most modules have detailed descriptions
- Most modules have 3+ key parts
- Most modules have risk matrices
- 8/8 modules have engineering reviews

BOM

Score: 9/10

EVALUATION REASONS

- 53 parts — comprehensive BOM
- Most parts have cost estimates
- Most parts have mass estimates
- Parts are linked to modules
- Total BOM cost: £3951

Cost

Score: 8/10

EVALUATION REASONS

- Unit cost: £5927
- Per-module breakdown available
- NRE estimated
- Arithmetic closes within 1%

Risks

Score: 8/10

EVALUATION REASONS

- 24 risks identified — comprehensive
- Most risks have mitigations
- Most risks have owners assigned
- 13 high-priority risks (score ≥ 10)

Suppliers

Score: 4/10

EVALUATION REASONS

- Most suppliers have match scores

IMPROVEMENT SUGGESTIONS

- Need more supplier matches
- More parts need supplier matches

Research

Score: 7/10

EVALUATION REASONS

- Research report: 11510 chars
- 10 sources cited
- 8 standard codes identified
- Domain detected: commercial_hvac